

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: DIGITAL SIGNAL AND IMAGE PROCESSING (01CT0513)	Aim: Perform FFT and IFFT on discrete time signal.	
Experiment -05	Date:	Enrollment No: 92410133069

Code :

```
import numpy as np
import matplotlib.pyplot as plt

# Discrete-time signal
signal = np.array([1, 2, 3, 4, 5, 6, 7, 8])

# Compute FFT
fft_result = np.fft.fft(signal)

# Compute magnitude spectrum
magnitude_spectrum = np.abs(fft_result)

# Compute phase spectrum
phase_spectrum = np.angle(fft_result)

# Compute IFFT
reconstructed_signal = np.fft.ifft(fft_result)

print("Original Signal:")
print(signal)

print("\nFFT Result:")
print(fft_result)
```

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```
print("\nMagnitude Spectrum:")
print(magnitude_spectrum)
```

```
print("\nPhase Spectrum:")
print(phase_spectrum)
```

```
print("\nReconstructed Signal using IFFT:")
print(reconstructed_signal)
```

```
# Plot results
plt.figure(figsize=(10, 8))
```

```
plt.subplot(3, 1, 1)
plt.stem(signal)
plt.title("Original Signal")
plt.xlabel("n")
plt.ylabel("Amplitude")
```

```
plt.subplot(3, 1, 2)
plt.stem(magnitude_spectrum)
plt.title("Magnitude Spectrum")
plt.xlabel("Frequency Index")
plt.ylabel("Magnitude")
```

```
plt.subplot(3, 1, 3)
plt.stem(phase_spectrum)
```

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```
plt.title("Phase Spectrum")
plt.xlabel("Frequency Index")
plt.ylabel("Phase (radians)")
```

```
plt.tight_layout()
plt.show()
```

Output :

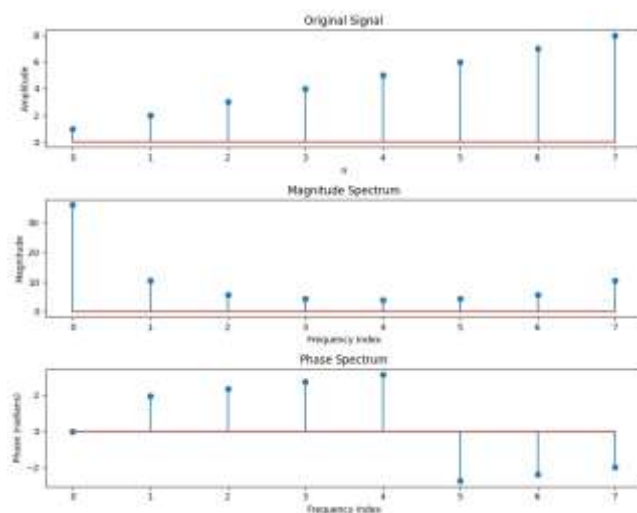
```
Original Signal:
[1 2 3 4 5 6 7 8]

FFT Result:
[36.+0.j          -4.+9.65685425j -4.+4.j          -4.+1.65685425j
 -4.+0.j          -4.-1.65685425j -4.-4.j          -4.-9.65685425j]

Magnitude Spectrum:
[36.  10.45258372  5.45685425  4.3295688  4.         4.3295688
  5.45685425 10.45258372]

Phase Spectrum:
[ 0.         1.96349341  2.35619449  2.74889357  3.14159265 -2.74889357
 -2.35619449 -1.96349341]

Reconstructed Signal using IFFT:
[1.+0.j  2.+0.j  3.+0.j  4.+0.j  5.+0.j  6.+0.j  7.+0.j  8.+0.j]
```



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Conclusion :

Thus, the FFT and IFFT of the given discrete-time signal were successfully performed using Python. The magnitude and phase spectrum were obtained, and the IFFT successfully reconstructed the original signal.

Github Link :

<https://github.com/Shashi-kumar-g/DSIP-LAB>
